

MSSV: 20235362

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BÀI TẬP NGÀY 28/04/2026 - LẬP TRÌNH MẠNG

Bài 1: Xây dựng chương trình subscribe_server sử dụng hàm select()/poll()

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <stdlib.h>
4 #include <unistd.h>
5 #include <arpa/inet.h>
6 #include <sys/socket.h>
7 #include <poll.h>
8
9 #define MAX_CLIENTS 100
10 #define BUFFER_SIZE 1024
11 #define MAX_TOPICS 100
12
13 struct Client {
14     int fd;
15     char topics[MAX_TOPICS][50];
16     int topic_count;
17 };
18
19 struct Client clients[MAX_CLIENTS];
20
21 struct pollfd fds[MAX_CLIENTS + 1];
22 int nfds = 0;
```

Terminal Output:

```
nb1@LAPTOP-72CTQ1VN:~/projects/Bef:
ore$ ./Subscribe_server_poll
Received: UNSUB animals
Client 4 unsubscribed animals
Received: SUB animals
Client 6 subscribed animals
Received: PUB animals Dog are cute too

nb1@LAPTOP-72CTQ1VN:~/projects/Bef:
ore$ nc localhost 9000
SUB animals
SUB weather
SUB sports
[sports]: Ronaldo
[animals]: Cats are cute
UNSUB animals

nb1@LAPTOP-72CTQ1VN:~/projects/Bef:
ore$ nc 127.0.0.1 9000
SUB animals
SUB weather
PUB animals Cats are cute
PUB animals Dog are cute too

nb1@LAPTOP-72CTQ1VN:~/projects/Bef:
ore$ nc 127.0.0.1 9000
SUB sports
PUB sports Ronaldo
SUB animals
[animals]: Dog are cute too
```

Hình 1. Minh họa chương trình với 3 clients